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Influencer Marketing on Instagram: Empirical Research on Social Media Engagement with Sponsored Posts

Jana Gross^a  and Florian von Wangenheim^b 

^aKEDGE Business School, Talence, France; ^bETH Zurich, Zurich, Switzerland

ABSTRACT

While sponsored posts by social media influencers (SMIs) on Instagram have dramatically increased as an advertising strategy, empirical results on their efficiency have yet to emerge. These sponsored posts are placed next to nonsponsored posts (NSPs) on SMIs' Instagram profiles. In the current study, we investigate how consumers engage with sponsored posts compared to NSPs and how engagement varies when such are posted by microinfluencers compared to macroinfluencers. Moreover, we explore engagement with sponsored posts containing different advertising appeals based on the advertising strategy, which determines whether sponsored posts contain appeals based on factual and rational information (i.e., informational appeals) or emotional and personal information (i.e., transformational appeals). Using a binomial regression on a unique data set of 64,438 sponsored Instagram posts by SMIs, we show that (1) users engage more with sponsored compared to NSPs, (2) particularly by microinfluencers, and (3) users engage more with sponsored posts with informational appeals by macroinfluencers than by microinfluencers. These findings provide meaningful insights for both advertisers interested in employing influencer marketing and SMIs interested in cooperating with advertisers for sponsored posts; the findings also advance the literature on influencer advertising and marketing.

KEYWORDS

Field data; influencer advertising; Instagram; social media analytics; social media influencers

In recent years, advertising scholars have continued to emphasize the crucial role of social media influencers (SMIs) in advertising (e.g., see Lou and Yuan 2019; Tafesse and Wood 2021). SMIs are individuals or groups of individuals who aggregate followers of their social media profiles (De Jans, Cauberghe, and Hudders 2018; Gross and von Wangenheim 2018). Advertisers choose and select SMIs for sponsored advertising based on social media engagement with their posts. Moreover, advertisers evaluate and reimburse SMIs based on the amount of social media engagement generated by their posts. As such, social media engagement is an important key performance indicator (KPI) to measure the success of sponsored posts on social media (Hughes, Swaminathan, and Brooks 2019; Lammenett 2019).

Sponsored posts are posts with a clear advertising message produced by SMIs and uploaded to their

social media profiles. In this fairly new form of advertising, advertisers employ SMIs to promote their brands and products on SMIs' social media profiles (De Veirman, Cauberghe, and Hudders 2017; Evans et al. 2017). Sponsored posts blur the line between paid and earned content, because advertisers reimburse SMIs for sponsored posts but SMIs have control over the sponsored posts' ultimate messages (Hughes, Swaminathan, and Brooks 2019). While users are likely to avoid or ignore traditional ads (Kelly, Kerr, and Drennan 2010), users recognize SMIs' posts as sponsored and willingly engage with them (Evans et al. 2017). In contrast, nonsponsored posts (NSPs) are posts without any advertising message or relation to a brand or product. Advertisers evaluate social media engagement of both sponsored posts and NSPs when considering cooperating with SMIs. As such, the field needs to develop a better understanding of

CONTACT Jana Gross  jana.gross@kedgebs.com  Department of Marketing, KEDGE Business School, 680 Cours de la Libération, Talence 33405, France.

Jana Gross (PhD, ETH Zurich) is an assistant professor, Department of Marketing, KEDGE Business School.

Florian von Wangenheim (PhD, University of Mainz) is a professor, Technology Marketing Group, and head, Department of Management, Technology, and Economics, ETH Zurich.

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whether and how these types of posts affect social media engagement.

To drive social media engagement of sponsored posts, advertisers might rely on informational and transformational appeals in sponsored posts as an advertising strategy. Informational posts contain information-based appeals that provide rational and useful information directly linked to the advertised brand or product (Puto and Wells 1984; Roose, Geuens, and Vermeir 2018). In contrast, transformational posts contain experience-based appeals (Puto and Wells 1984; Roose, Geuens, and Vermeir 2018) and provide emotional and personal experiences about the advertised brand or product. Both appeals positively affect advertising attitude, purchase intentions, and consumer experience (Akbari 2015; Naylor et al. 2008), yet they are employed to accomplish different advertising goals. In the context of SMIs, advertisers employ informational appeals in sponsored posts to provide immediate benefit to consumers that require immediate actions, as such, designed to influence short-term consumer engagement and actions (i.e., sales, leads, or brand awareness). This is in line with Grigaliunaite and Pileliene (2016), who found that informational appeals attract more attention from customers compared to transformational appeals. Because transformational appeals operate mainly on experiences, emotions, and feelings (Puto and Wells 1984), these appeals are employed to produce superior long-term consumer engagement and actions (i.e., brand attitude, brand loyalty). Indeed, transformational appeals have been found to positively affect brand liking, brand knowledge, brand attitude, and word of mouth (WOM) (Naylor et al. 2008; Teichert, Hardeck, Liu, and Trivedi 2018). Despite this promising advertising strategy in practice, it remains unexplored whether and how informational and transformational appeals affect engagement with sponsored posts. Given the significant marketing expenditures dedicated to this method, the strategy of using different appeals to drive engagement with sponsored posts is an important research gap worth addressing for scholars, advertisers, and SMIs.

To investigate the effect of advertising appeals on social media engagement with sponsored posts, it is crucial to understand how these effects vary between microinfluencers and macroinfluencers. SMIs build solid bases of followers by producing and uploading posts to their social media profiles (Yuan and Lou 2020). SMIs with more followers are referred to as macroinfluencers, and SMIs with fewer followers are referred to as microinfluencers (Voorveld 2019).

Despite the higher public exposure and reach of sponsored posts by macroinfluencers (De Veirman, Cauberghe, and Hudders 2017), Brewster and Lyu (2020) found that parasocial relationships and interactions are significantly higher for SMIs with fewer followers. Building on the elaboration likelihood model (ELM; Cacioppo and Petty 1984; Petty and Cacioppo 1986), advertising messages by macroinfluencers are likely to be processed more attentively and, thus, rely less on heuristics, such as the parasocial interaction. In line with this reasoning, research has found that microinfluencers positively affect engagement (Marques, Casais, and Anthony Camilleri 2021). However, research has yet to explore whether and how SMIs' follower counts influence the effect of different advertising appeals (i.e., informational versus transformational) on engagement with sponsored posts.

Despite the existence of many studies investigating the effects of company-generated and consumer-generated content on engagement (Cvijikj and Michahelles 2013; De Vries, Gensler, and Leeflang 2012; Sabate et al. 2014), the literature does not closely consider sponsored posts by SMIs—in other words, SMI-generated content on social media. Moreover, although there have been some recent studies on influencer marketing (Eisend et al. 2020; Evans et al. 2017; Lou and Yuan 2019; Yuan and Lou 2020), none of these studies focused on the effectiveness of sponsored posts. In the current study, we seek to fill this research gap. The objectives of this article are threefold: First, this study provides insights regarding how social media users engage with sponsored posts compared to NSPs in general. Second, this research then explores the effect of advertising appeals on the effectiveness of sponsored posts in particular. Finally, building on the previous objectives, the current study investigates how SMIs' follower count influences the effects of advertising appeals on engagement with sponsored posts.

This article provides novel insights for both advertisers and scholars. It empirically tested the previously mentioned research objectives using 64,438 Instagram posts from SMIs. Our field data provide a unique perspective and draw a richer picture of which drivers affect the success of influencer marketing and how they vary with SMIs' follower count. From a practitioner perspective, this study provides insights into a deeper understanding to advertisers on how users engage with SMI-generated posts in general and sponsored posts in particular. The findings have implications for practitioners who want to employ SMIs and

show that the advertising appeal in sponsored posts should be guided by type of SMIs.

From a research perspective, our study has important theoretical contributions in two ways. In a narrow sense, we contribute to advertising and influencer literature. We extend research incorporating endorsers in advertising in social media (e.g., see Carrillat and Ilicic 2019; Schouten, Janssen, and Verspaget 2020). We show that users engage more with sponsored posts than NSPs, suggesting that employing SMIs and their sponsored posts is an effective advertising strategy on Instagram. In a broader sense, we provide first findings that users engage more with sponsored posts by microinfluencers, particularly when they use transformational appeals. That is, our study employs information processing theory to specify how type of advertising message influences engagement with sponsored posts by micro-compared and macroinfluencers.

Theoretical Background

Nonsponsored Posts by Social Media Influencers

SMIs build marketing value to advertisers by regularly producing content and cultivating a sizable number of followers (Lou and Yuan 2019). SMIs build up their marketing value and establish themselves by producing and uploading NSPs. These posts are unrelated to a company and do not refer to any particular brands or products. They contain stories and anecdotes of SMIs' lives to attract the attention of followers. Their messages do not contain any form of commercial or branded message. As such, SMIs present a personal and editorial style through these posts on their Instagram profiles. Advertisers evaluate NSPs when considering SMIs for sponsored posts, which present important marketing value to advertisers.

NSPs should not be mistaken for organic consumer communication. We distinguish NSPs from organically generated WOM because SMIs do not talk about and relate to brands and products. As SMIs do not engage in WOM (neither in the caption nor in the image), NSPs are purely editorial posts. Because NSPs by SMIs are unrelated to brands and products, we can distinguish this study from prior research focusing on WOM, including user-generated content (Berger 2014; Dunn and Harness 2019) or native advertising (Wojdyski 2016).

Sponsored Posts by Social Media Influencers

Similar to NSPs, sponsored posts are produced by SMIs and are uploaded to their social media profiles.

Although SMIs produce both sponsored posts and NSPs, they differ in the information they provide related to brands and products. In contrast to NSPs, sponsored posts contain a clear promotional or advertising message about brands and products (Boerman, Willemsen, and Van Der Aa 2017). Sponsored content (or posts) refers to the integration of brands and branded messages into editorial content and compensated by a sponsor (Eisend et al. 2020; van Reijmersdal et al. 2020). Sponsored posts have been incentivized and influenced by companies (Hughes, Swaminathan, and Brooks 2019). As such, companies have some creative control to ensure that SMIs stay on the advertising message (Childers, Lemon, and Hoy 2018).

Sponsored posts differ from NSPs in that they have a caption with a clear advertising message. Both sponsored posts and NSPs usually involve images and captions. Captions are supplemental textual information added to an image and reflect an image's message. The alignment between images and captions or text positively affects attitudes toward products (Van Rompay, De Vries, and Van Venrooij 2010). Companies use the captions of sponsored posts as an advertising tool; they contain advertising messages and provide useful information about brands and products, such as where to buy a product, where to get the best offers, or an SMI's experience of a product. These advertising messages may help users save money and make better purchase decisions.

Advertising Appeals in Sponsored Posts

Advertisers employ two advertising appeals in sponsored posts. The first is known as informational advertising in the literature (Roose, Geuens, and Vermeir 2018; Rossiter, Percy, and Bergqvist 2018). We follow the definition of Puto and Wells (1984) who define informational advertising as "appeals providing consumers with factual, relevant brand data in a clear and logical manner such that they have greater confidence in their ability to assess the merits of buying the brand after having seen the advertisement." Informational advertising is based on providing rational information that is directly linked to the advertised brands and products (Roose, Geuens, and Vermeir 2018). The characteristics of informational advertising are that it contains factual, relevant information about brands and products; information that is immediately and obviously important to consumers; and data that are perceived by the consumer as verifiable (Puto and Wells 1984). The goal of informational appeals is to focus on features or benefits of the brand

or product itself in the advertising message (Puto and Wells 1984).

In the context of sponsored posts, advertisers use informational messages to provide immediate important and relevant information about brands and products to consumers. These posts contain advertising appeals designed to generate specific short-term responses and engagement from consumers. The goal of these posts is to sway consumers into immediate action. This type of advertising appeal is a short-term tactic designed to result in immediate benefits and effects for advertisers, such as awareness, leads, or sales (Funke 2019). For example, a commonly used appeal for such information in sponsored posts is a discount or promotional code provided in a caption (Understanding Ecommerce 2020). The idea is that consumers can benefit from these discount codes within a certain time frame with the goal to push immediate sales for the advertiser. We refer to this type as a sponsored post with informational appeal.

The second type of advertising appeal is referred to as transformational appeal in the literature (Roose, Geuens, and Vermeir 2018; Rossiter, Percy, and Bergvist 2018). Puto and Wells (1984) define transformational advertising as “appeals that associate the experience of consuming or using the advertised brand with a unique set of psychological characteristics which would not typically be associated with the brand experience to the same degree without exposure to advertising” (p. 638–643). Transformational advertising is based on providing information that is imaginary based and links experiences and feelings to the advertised brands and products (Roose, Geuens, and Vermeir 2018). Transformational advertising contains the experience of using the brand or product as richer, warmer, more exciting, and/or more enjoyable than that obtained solely from an objective description of the advertised brand (Puto and Wells 1984). Transformational appeals focus on showing the positive experiences of using the advertised brands or products (Naylor et al. 2008).

In the context of sponsored posts, advertisers use transformational messages to provide experience-based personal information about brands and products to consumers. These posts contain advertising appeals designed to generate long-term responses and engagement from consumers. This type of advertising appeal is a long-term tactic, designed to result in lasting effects for the advertisers, such as improved brand image or brand loyalty. The goal of these posts is to encourage and strengthen consumers’ perception of brands and products. For example, a commonly used appeal for

such information in sponsored posts involves SMIs recommending their favorite flavors or colors of a new product, which is usually provided in captions. The idea of these type of posts is to establish and foster the benefits of the advertised brand or product in the minds of the audiences of the SMIs. We refer to this type as a sponsored post with transformational appeal.

While both types of messages can occur in the same post, one of the approaches will dominate (Cutler, Thomas, and Rao 2008) in practice. Companies only partly control the advertising message because SMIs have the ultimate control of the messages they communicate (Hughes, Swaminathan, and Brooks 2019). Hence, advertisers can guide SMIs toward informational messages by providing incentives that contain factual and relevant information with immediate benefit to consumers. To do so, advertisers use promotional incentives that call consumers to action. Typical examples are discount codes or giveaways. SMIs then embed these promotional incentives into their advertising messages. Therefore, SMIs either use informational or transformational appeals in their advertising messages in sponsored posts. In the context of this research, we distinguish between informational and transformational messages in sponsored posts.

Social Media Engagement As Key Performance Indicator for Influencer Marketing

Consumer engagement on social media refers to “an expression of consumers’ cognitive and emotional attitudes via their brand-related engagement behaviors in social media” (Pentina, Guilloux, and Micu 2018, p. 57). Following this definition, engagement on social media includes both behavioral and attitudinal actions by consumers toward brands. In line, social media engagement is a set of brand-related actions by consumers toward brands, products, or services. Regarding influencer marketing, SMIs produce posts sponsored by companies and upload them to their Instagram profiles, and consumers may interact with the company by clicking the Like button on the post or by writing a comment about the post. Muntinga, Moorman, and Smit (2011) identified clicking the Like button or writing a comment below content as brand-related actions on social media. True engagement on social media is when consumers feel passionately enough to become active, take action, and interact (Yoon et al. 2018). Within influencer marketing, liking and commenting is characterized as profound and true engagement.

While liking and commenting can be conceptually regarded as digital and consumer engagement (Rodgers and Thorson 2018; Thorson and Rodgers 2006; Yoon et al. 2018), the level of engagement expressed with these social media actions is highly dependent on context platform. Instagram has been found to be used mainly for social interaction (Voorveld et al. 2018). Users engage with Instagram to pass the time, fill an empty moment, and interact with others (Voorveld et al. 2018). Users are looking for interaction and engagement with others on Instagram. Hence, posts that generate likes and comments reflect a high level of social media engagement for companies in the context of influencer marketing.

Likes and comments are KPIs in influencer marketing (Hughes, Swaminathan, and Brooks 2019). Advertisers evaluate the performance of sponsored posts based on total social media interactions. Hence, advertisers pay SMIs based on the number of likes and comments generated by their sponsored posts. Moreover, likes and comments are significant predictors of companies' financial and social performance, and they positively influence sales (Lee, Lee, and Oh 2015; Yoon et al. 2018); beyond financial success, they positively affect brand trust, involvement, and overall company reputation (Ji et al. 2017; Phua and Ahn 2016). Moreover, developing favorable message evaluations leads to greater online engagement (Alhabash et al. 2015). Companies may consider likes and comments not only as a return on investment in social media environments but also the total number of likes of and comments on sponsored posts as consumer engagement with their brands and products. Therefore, we use the number of likes and comments as a proxy for social media engagement.

Linking Sponsored and Nonsponsored Posts to Social Media Engagement

To provide a priori evidence of the differences between sponsored posts and NSPs, we conducted a qualitative prestudy. We interviewed six professional Instagram SMIs ($N=6$, $M_{age} = 30$, 83.33% male). We chose SMIs as interview partners because they provide a diverse perspective due to their dual roles as producers and consumers of both sponsored posts and NSPs. The participants were highly experienced in cooperating with companies for sponsored posts on Instagram. Of the interviewees, two are full-time professional SMIs; the remaining are half-time professional SMIs. All interview partners are highly experienced and have cooperated with a wide range of

companies for sponsored posts. Each interview was semistructured and lasted between 30 and 60 minutes and was audio recorded. The participants were between 20 and 35 years old. They discussed their experiences and choices of companies for sponsored posts, as well as their production processes from idea development to upload of sponsored posts and NSPs.

Comparing the production process between sponsored posts and NSPs, they reported two differences. First, SMIs invest more time, thought, and effort into producing sponsored posts than nonsponsored ones because they are financially reimbursed by advertisers for sponsored posts. Because advertisers pay SMIs for producing and uploading sponsored posts, they aim to satisfy companies by producing creative sponsored posts. Interview partners state:

I think much more about how I implement a sponsored post [compared to a nonsponsored post] [. . .] the entire creative process and the preparation process [of sponsored posts] is longer.

If companies contact me and want to do something with me, then I want to produce [a sponsored post] that benefits them. [. . .] It is a giving and a taking.

Second, SMIs feel creatively inspired and challenged to embed the advertising appeal in a credible and compelling manner into their posts. They argue that advertisers show trust toward SMIs by allowing them to incorporate the advertising appeals in ways that the SMIs believe works best for their audiences. In turn, SMIs want to reciprocate this trust by showing their ability to produce posts that satisfy both the audiences' and advertisers' needs and preferences. Interview partners claim:

You always have some requirements when cooperating with a [company]. Working around it promotes my creativity. It doesn't inhibit me.

The company's requirements broaden my creative horizon. [. . .] I prefer a few requirements to no requirements at all or too many requirements.

The interviews revealed that SMIs invest more energy into producing sponsored posts while being challenged to embed advertising appeals in a creating and compelling manner. As such, SMIs (in their role as producers) provide and (in their role as consumers) receive richer content in sponsored posts compared to nonsponsored ones. In this vein, SMIs claimed that sponsored posts are more interesting and useful compared to nonsponsored ones, because they offer additional value to consumers. Sponsored posts contain information about brands and products which nonsponsored ones do not. As a result, findings from the interviews indicate that social media users experience

a higher hedonic value when exposed to sponsored posts compared to nonsponsored ones. Interview partners said:

The audience has a clear added value with sponsored posts. They learn something about a product, an adventure they can have, they see new trends, they discover something.

[In sponsored posts] the consumer has an added value. I think [as a consumer and SMI] it's cool to get innovation, new products, and simply the information [in sponsored posts]. I appreciate that.

It is likely that hedonic value is higher for sponsored posts than for NSPs. The hedonic value of a post refers to the enjoyment, vividness, and entertainment a user experiences from reading the post (Hughes, Swaminathan, and Brooks 2019). Social media platforms like Instagram have been identified as entertainment-oriented environments (Schulze, Schöler, and Skiera 2014). Hence, users are looking for hedonic value when being exposed to Instagram posts. Sponsored posts contain a clear promotional or advertising message about brands and products (Boerman, Willemsen, and Van Der Aa 2017). NSPs are unrelated to a company and do not refer to any particular brand and product. We argue that the added hedonic value in sponsored posts results from the advertising messages about brands and products. Therefore, these results lend support to our argument that hedonic value differs between sponsored posts and NSPs, with hedonic value being higher for sponsored posts.

Previous research suggest that hedonic value can have an impact on post interaction and popularity (De Vries, Gensler, and Leeftang 2012; Hughes, Swaminathan, and Brooks 2019; Sabate et al. 2014). In the context of company-generated brand posts, De Vries, Gensler, and Leeftang (2012) found that brand posts with higher level of vividness are more popular. Similarly, Sabate et al. (2014) found that vividness positively affects interaction with brand posts. Relatedly, a study by Luarn, Lin, and Chiu (2015) showed that entertaining brand posts positively affect likes and comments of a post more than nonentertaining brand posts do. Extending those findings, Muntinga, Moorman, and Smit (2011) identified entertainment as a main motivation for contributing to brand-related content. Building on these findings, we assume that users experience higher hedonic value from sponsored posts, which results in more social media engagement, compared to NSPs. We propose:

H1: Users engage more with sponsored posts than with nonsponsored posts.

On social media platforms, SMIs' follower counts reflects their degree of public exposure and reach. In contrast to macroinfluencers, microinfluencers are perceived as more relatable and trustworthy due to their lower follower count (Britt et al. 2020). While sponsored posts by macroinfluencers are perceived as less authentic and more biased, those by microinfluencers are perceived as more authentic and honest (Gross 2020). Moreover, Lou and Yuan (2019) found that SMIs' trustworthiness positively affects followers' trust in sponsored posts. One can expect that social media engagement with sponsored posts varies between micro- and macroinfluencers.

Compared to microinfluencers, macroinfluencers' sponsored posts are exposed to more users. The number of followers is an indicator of popularity and reflects an SMI's network and group size (De Veirman, Cauberghe, and Hudders 2017). Darley and Latané (1968) showed that people in larger groups feel less responsible and less personally addressed, which creates a bystander effect. Building on these findings in the offline environment, the same may be true for social media environments such as Instagram or YouTube. Users may feel less responsible and addressed by macroinfluencers' sponsored posts compared to those by microinfluencers, resulting in a looking effect. In line with this argument, Brewster and Lyu (2020) showed that parasocial relationship and interaction is significantly higher for SMIs with fewer followers. Relatedly, recent research has found microinfluencers to positively affect engagement (Marques, Casais, and Anthony Camilleri 2021). Similarly, Tafesse and Wood (2021) showed that number of followers is negatively associated with engagement on social media. Hence, users may engage passively with macroinfluencers' sponsored posts (i.e., looking at them), rather than engaging actively with them (i.e., liking or commenting). We propose:

H2: The number of followers weakens the relationship between social media engagement with sponsored posts compared to nonsponsored posts.

Because one of our research goals is to understand if and how informational and transformational appeals affect engagement with sponsored posts subsequently, we focus only on sponsored posts for the following hypotheses. When including advertising messages in their posts, SMIs produce and upload sponsored posts to their Instagram profiles. While advertisers evaluate NSPs when considering cooperating with SMIs, advertisers pay SMIs to embed advertising messages into their sponsored posts. Sponsored posts come with two

types of advertising appeals: (1) informational appeals and (2) transformational appeals. While we expect engagement to vary between sponsored and NSPs, we also expect the type of advertising appeal in sponsored posts to affect engagement.

Sponsored posts with informational appeals contain information about brands and products that directly and immediately provide benefit to users. These incentives are advertising strategies designed to elicit specific responses and engagement (Hughes, Swaminathan, and Brooks 2019). These strategies aim to increase immediate customer engagement, the desire to buy more, lead generations, or conversions (Hughes, Swaminathan, and Brooks 2019; Verhoef, Reinartz, and Krafft 2010). In contrast, transformational appeals contain information about brands and products that are linked to personal experiences or stories. We can think of sponsored posts with transformational appeals as company-initiated recommendations because advertisers initiate and engineer brand-related conversations (Godes and Mayzlin 2009). These strategies aim to increase long-term customer engagement, actions, and preferences (Berger 2014; Godes and Mayzlin 2009). Therefore, it is likely that users' engagement with sponsored posts varies depending on the advertising appeal of the sponsored posts.

Informational appeals in sponsored posts are incentive forms including informational value to users that call to action. These incentives prompt users to participate, do something, and become active. Prior research suggest that users might tend to engage more with informational appeals compared to transformational appeals. For example, Schultz (2017) found that number of likes is significantly positively affected by calls to action in brand messages. Similarly, Hughes, Swaminathan, and Brooks (2019) found that Facebook influencer campaigns that included incentives significantly increased the number of post comments. In line, Lou and Yuan (2019) found that the informational value of influencer-generated content positively affected trust in it. Building on those findings, we expect the type of advertising appeal to affect social media engagement with sponsored posts. It is reasonable to assume that users positively evaluate informational appeals due to the informational value in sponsored posts which, in turn, increases engagement compared to those with transformational appeals. We predict:

H3: Users engage more with sponsored posts with informational appeals than with sponsored posts with transformational appeals.

The ELM describes how mental resources affect consumers' information processing (e.g., see Petty and

Cacioppo 1986). Information processing theory suggests that consumers invest more attention, thoughts, and efforts in the advertising message when processing information by the central route (Cacioppo and Petty 1984; Petty, Cacioppo, and Schumann 1983). In contrast, consumers rely more on heuristics, social cues, and signals of a advertising message when processing information by the peripheral route. Hence, consumers may pay less attention to elaborating the advertising message and instead may react to the message based on signals and cues in the message, such as discount codes.

Social media platforms represent low-involvement, high-distraction, and hedonic environments (Hughes, Swaminathan, and Brooks 2019; Schulze, Schöler, and Skiera 2014). Instagram is an example of such a platform. In low-involvement environments, users put a greater emphasis on peripheral cues, such as the number of followers (Hughes, Swaminathan, and Brooks 2019). Despite feeling less responsible and less addressed by macroinfluencers' sponsored posts in general, users evaluate sponsored posts that are followed by more people as more interesting and relevant. Thus, they pay more attention to such posts when evaluating their advertising messages compared to those by microinfluencers. When users pay more attention to advertising, they put more mental effort into evaluating the message, and they do so more carefully (Cacioppo and Petty 1979; Petty, Cacioppo, and Heesacker 1981; Petty, Cacioppo, and Schumann 1983). When users pay less attention to advertising, they put less mental effort into evaluating the message, and they rely more on heuristics, signals, and cues (Petty, Cacioppo, and Heesacker 1981; Tellis 2003). Thus, social media engagement with sponsored posts with informational appeals should vary between micro- and macroinfluencers.

Sponsored posts with informational appeals contain incentives that immediately benefit users. This is not the case for sponsored posts with transformational appeals. These incentives promote information about advantages related to brands and products that are helpful for users and include a call to action (e.g., discount codes, giveaways, promotions). In line with predictions of the ELM, these incentives should exert a greater impact on persuasion when evaluating advertising messages more carefully; this implies that informational appeals should play an important role in eliciting social media engagement between micro- and macroinfluencers. We expect a higher number of followers to positively affect social media engagement with informational appeals compared to

Table 1. Sponsorship labels: Three perspectives on labeling sponsored posts.

Perspective	Labels
Researcher	#ad, #advert, #anzeige, #werbung, #paidpartnership, #sponsored, #collab, #advertisement, #paid, #paidad, #promotion, #promo, #promotional, #branded, #partner ad by, product placement, advertisement, [werbung], [advert], [anzeige], ad, advert, anzeige, werbung, thanks to, sponsored, paid, promotion, sponsored by, collaboration, provided by
Company	#werbung, #bezahltepartnerschaft, #ad, #paidpartnership
Social media influencer	anzeige, ad, #ad, #sponsored, #werbung, #sponsoredby, #advert, bezahlte partnerschaft mit, [werbung]

transformational appeals in sponsored posts. We therefore propose:

H4: The number of followers strengthens the relationship between social media engagement with sponsored posts with informational appeals compared to those with transformational appeals.

Method

To test the proposed hypotheses and answer the research objectives, we adopted a field data approach. We collected data from an influencer marketing agency that helps companies recruit SMIs for sponsored posts. SMIs produce and upload sponsored posts to their social media profiles, embedding advertising messages in the captions of sponsored posts. The agency focuses on SMIs in sports, lifestyle, and health. We analyzed a set of 64,438 Instagram posts.

Measurements of the Independent Variables

Identification of Sponsorship in Instagram Posts

To operationalize the type of post (i.e., sponsored versus nonsponsored), we built a sponsorship label list. Sponsorships are labeled in the captions of Instagram posts. While labeling sponsored posts is mandatory, there are no strict rules or fixed regulations regarding the labels that must be used to disclose sponsorship. Because disclosing sponsorships is mandatory on social media, neglecting to do so might result in legal proceedings. Therefore, SMIs are interested and recommended to label their sponsored posts accordingly.

To build a comprehensive sponsorship labels list, we conducted three prestudies. The goal of the prestudies was to account for as many different labels as possible in the list. To create a comprehensive sponsorship label list, we conducted each prestudy from a different perspective. First, we conducted observational research with 50 full-time SMIs. We observed 10 to 20 posts per SMI and manually collected all the different labels and label styles they used for their sponsored posts. Second, we collected information from an influencer marketing agency about (1) which labels they recommend SMIs use when labeling their

posts as sponsored and (2) their experiences regarding which labels are most often used by SMIs. Third, we surveyed seven full-time SMIs about their sponsorship labels in their captions. We asked them to provide the three labels they use most often to label their sponsored Instagram posts.

Results of the three prestudies showed that sponsored posts are mainly labeled at the beginning of the caption. SMIs use labels either with or without a hashtag. Agencies provide several labels to SMIs, show examples of how to label posts, and recommend doing so at the beginning of the caption. Many SMIs use both the label in the beginning of the caption and add a label in form of a hashtag at the end of the caption. For our sample, SMIs labeled sponsored posts either in German (e.g., *Werbung*) and/or in English (e.g., *advert*). Results also showed that SMIs care about being transparent and honest with their followers about their sponsorship and label their posts properly.

We collected and combined all labels from the three prestudies to build a sponsorship labels list. To identify the sponsored posts in the data set, we used the labels reported in Table 1. We used a text-analytical approach to reliably quantify whether a post was sponsored in the caption. We performed word detection to identify whether the caption contained one of the sponsorship labels in the list.

To ensure that the list does correctly detect sponsored posts in our sample, we performed a manual face validity check of a subsample of 2,000 sponsored posts and NSPs each to check and control the accuracy of the identification of the type of post. We could not detect any error in the label detection. We operationalized sponsorship as a binary variable: SPONSOR = 1 if the caption of the Instagram post contained a sponsorship label, and 0 otherwise.

Identification of Advertising Appeal in Sponsored Posts

Although Puto and Wells (1984) argued that ads can contain both informational and transformational appeals, there are difficulties with such implementation in practice. It is possible for sponsored posts to contain informational and transformational appeals, yet these appeals are mostly used independently. SMIs

have control of the ultimate advertising message to be communicated in sponsored posts (Hughes, Swaminathan, and Brooks 2019). As a result, companies only partly control the message of the caption by defining the goal of the sponsored posts. They can guide SMIs into using either informational or transformational appeals when advertising brands and products. In line with this, Cutler, Thomas, and Rao (2008) argued that, in practice, one appeal is more dominant than the other.

Companies employ influencer marketing and cooperate with SMIs for sponsored posts with different goals (Funke 2019). Companies differentiate two major types of sponsored posts: Those that aim to increase long-term consumer actions (e.g., brand awareness, loyalty, and image) and those that aim to increase short-term consumer actions (e.g., sales, leads, and traffic). To increase short-term and immediate action by consumers, companies offer SMIs promotional incentives with informational value. SMIs then embed these promotional incentives into their sponsored posts. Typical examples of promotional incentives that aim to increase short-term and immediate action by consumers are discount codes, links to online stores, challenges, or giveaways (Funke 2019). Figure 1 presents an example of a caption of a sponsored post that contains a discount code as promotional incentive.

According to Puto and Wells (1984), for an ad to be informational it must contain the following characteristics: (1) It must “present factual, relevant

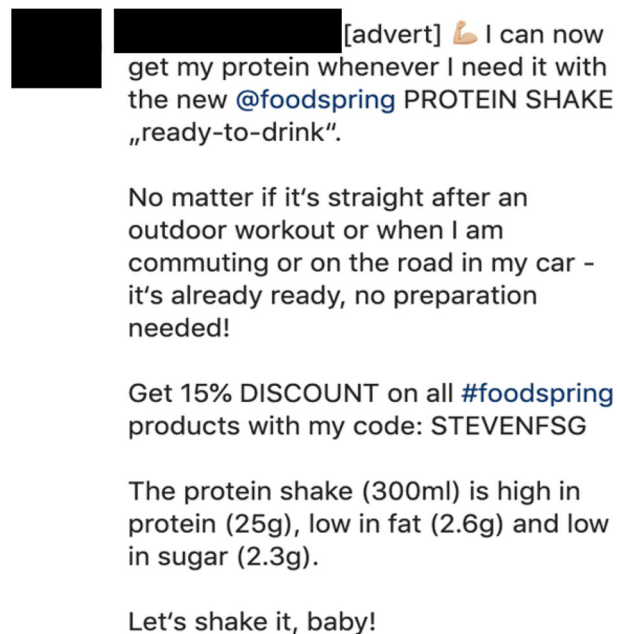


Figure 1. Example of a caption of a sponsored post with a discount code (source: Instagram, 2022).

information about the brand,” (2) “present information which is immediately and obviously important to the potential consumer,” and (3) “present data which the consumer accepts as being verifiable.” The discount code used in the caption presented on Figure 1 contains all three characteristics. First, the discount code is objective and relevant information because it presents the reduction in price when buying the product. Second, the discount code presents information that is immediately and obviously important to potential consumers, because discount codes are valid only for a certain amount of time. Finally, discount codes can be verified by consumers when using them at purchase. Hence, these promotional incentives represent informational appeals in sponsored posts to consumers.

In the present study, we compare sponsored posts with informational appeals to those with transformational appeals. We chose to operationalize the advertising appeal as a binary variable (INFO) that equals 1 for informational appeal and 0 for transformational appeal in sponsored posts. A second approach to operationalize the advertising appeal in sponsored posts would be to use a continuous variable indicating the degree of each appeal used in a post (e.g., 20% informational and 80% transformational). Another alternative approach to operationalize the advertising appeal in sponsored posts would be to use a categorical variable ranging from 1 to 4 and indicating the types of appeals used in sponsored posts (e.g., 1 = *Informational*, 2 = *Transformational*, 3 = *No appeal*, 4 = *Informational and transformational*). Both alternative operationalizations would require companies, agencies, and SMIs to employ both types of appeals in sponsored posts.

While advertising appeals are not mutually exclusive categories of advertising, the weight of informational and transformational appeals in advertising messages varies (Teichert et al. 2018). Both types of appeals can occur in the same advertising message, but one of them will dominate (Cutler, Thomas, and Rao 2008). Informational appeals dominate when the advertising is focusing on brands and products themselves (Puto and Wells 1984). In contrast, transformational appeals dominate when the advertising is focusing on the experience of brands and products (Puto and Wells 1984). Thus, an advertising message is likely to focus on one advertising appeal depending on its goal. This reasoning is in line with previous research that used a dichotomous classification to categorize informational and transformational appeals (e.g., see Akbari 2015; Laskey, Day, and Crask 1989;

Roose, Geuens, and Vermeir 2018). Therefore, we chose a dichotomous classification of the advertising appeal.

A dichotomous classification of the advertising appeal is in line with practices in the influencer industry. Companies provide promotional incentives as informational appeals, which SMIs, in turn, embed into their advertising messages in sponsored posts. These posts are then thematically focused on these informational appeals, which exclude transformational appeals in the same post. In our study context, informational and transformational appeals are used independently. We follow this practice and categorize advertising appeals as either informational or transformational appeals in sponsored posts in our study. Therefore, in our analysis we rule out a continuous or categorical measurement for the advertising appeal.

The marketing agency provided information about the different promotional incentives SMIs use in the captions of their posts when informational appeals are used in sponsored posts. Based on the promotional incentives, we built a list of keywords and used it to identify if a sponsored post contained an informational appeal. Through word detection, we identified if a sponsored post contained one of the keywords in the informational appeal list. The list of keywords used to detect the type of advertising appeal in sponsored posts appears in Table 2.

We performed two manual face validity checks to ensure the reliability of the distinction between informational and transformational appeals. First, we manually controlled more than 2,000 transformational and 2,000 informational posts to ensure that the type of advertising appeal used in sponsored posts was not falsely assigned through word detection. We could not detect any errors in keyword detection. Second, Puto and Wells (1984) state that transformational advertising is affect based rather than cognitive based. As such, we would expect the number of characters in captions to vary among the different advertising appeals. We expect sponsored posts with informational appeals to have more characters because it takes more words to provide objective, factual, and relevant information about brands and products to users. In line with this argument, sponsored posts with informational appeals contain a mean of 439 characters compared to sponsored posts with transformational appeals with a mean of 355 characters.

Model Specification

We estimated the model as follows:

$$\begin{aligned} \text{ENGAGEMENT}_i = & \text{CONSTANT} + \alpha_1 \text{SPONSOR}_i \\ & + (\alpha_1 \text{INFO}_i) + \alpha_2 \text{FOL}_i \\ & + \alpha_3 \text{FOL}_i \text{SPONSOR}_i \\ & + (\alpha_3 \text{FOL}_i \text{INFO}_i) + \alpha_4 \text{TEXT}_i \\ & + \alpha_5 \text{EMOJI}_i + \alpha_6 \text{MENTION}_i \\ & + \alpha_7 \text{HASHTAG}_i + \alpha_8 \text{EXCL}_i \\ & + \alpha_9 \text{QUE}_i + \alpha_{10} \text{CONTENT}_i \\ & + \alpha_{11} \text{WORKHOUR}_i \\ & + \alpha_{12} \text{WEEKEND}_i + \varepsilon_i, \end{aligned}$$

where i indexes the post that has been uploaded to an Instagram profile ($i = 1, 2, \dots, 64,438$).

The dependent variable (DV) is the total engagement with a post (ENGAGEMENT), measured as the sum of the number of likes of and comments on a post. We use total engagement as DV because the total number of likes and comments serves as a pricing scheme for companies and SMIs.

On the right-hand side of the equation, the model consists of a sponsorship variable (SPONSOR) as a dummy variable that equals 1 if the post was labeled as sponsored and 0 otherwise. The appeal dummy variable (INFO) reflects the advertising appeal in sponsored posts that equals 1 for informational appeal and 0 for transformational appeal. We included a continuous variable for the number of followers of the SMI (FOL) who uploaded the post.

Besides the explanatory variables, we incorporated a few covariates into our model to rule out alternative explanations of the modeling results. First, social media activities and engagement might vary between weekend and weekdays. It may be that users visit SMI profiles more during the week than on weekends because they spend more time with leisure activities, friends, and families on weekends, resulting in higher engagement for posts uploaded during the week. This is in line with previous research that showed Internet search and social media activities are higher during the week than on weekends (De Vries, Gensler, and Leeflang 2012; Rutz and Bucklin 2011). Therefore, we included two indicator variables in the model to control for the differences in engagement by weekday and time. We added a weekend dummy variable

Table 2. Informational appeals in sponsored posts: Keywords.

Keywords

discount, promotion, giveaway(s), bonus, code, rabatt, gewinn, gwin, gwin, challenge, % off, gutschein, verlosung, save, CHF, gewinnspiel, win, voucher, teilnahmebedingungen, concours, contest, aktion

Table 3. Correlation matrix among the variables.

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1 LIKES	1														
2 COMMENTS	0.51	1													
3 ENGAGEMENT	1	0.51	1												
4 SPONSOR	0.04	0.17	0.04	1											
5 INFO	0.01	0.09	0.01	0.41	1										
6 FOL	0.76	0.51	0.76	0.08	0.02	1									
7 TEXT	-0.03	0.09	-0.03	0.17	0.19	0.02	1								
8 EMOJI	-0.01	0.06	-0.01	0.07	0.07	0.02	0.23	1							
9 MENTION	0.00	-0.03	0.00	0.07	0.07	0.01	0.17	0.1	1						
10 HASHTAG	-0.07	-0.09	-0.07	0.03	0.04	-0.08	0.48	0.00	0.00	1					
11 EXCL	-0.01	0.03	-0.01	0.06	0.08	0.03	0.29	0.12	0.13	0.00	1				
12 QUE	0.01	0.13	0.01	0.07	0.07	0.05	0.33	0.13	-0.03	0.05	0.11	1			
13 CONTENT	0.02	-0.01	0.02	-0.06	-0.02	0.05	-0.07	0.05	0.12	-0.06	0.03	-0.02	1		
14 WORKHOUR	-0.01	-0.01	-0.01	0.00	0.02	0.00	0.05	0.02	0.00	0.05	0.00	0.03	-0.01	1	
15 WEEKEND	0.00	-0.02	0.00	-0.01	-0.01	0.00	-0.01	-0.01	0.00	0.01	0.01	-0.01	0.00	0.05	1

(WEEKEND) that equals 1 if the post was uploaded on Saturday or Sunday and 0 otherwise. We also added a working hour dummy variable (WORKHOUR) that equals 1 if the post was uploaded from 8:00 a.m. to 6:00 p.m. and 0 otherwise.

Second, Instagram posts allow for captions of a maximum of 2,200 characteristics. These characteristics can be used for plain text, hashtags, mentions, or emojis and might positively or negatively affect engagement. In line with this, post characteristics have been found to affect engagement (De Vries, Gensler, and Leeflang 2017; Sabate et al. 2014). Moreover, previous research found that message length may either positively or negatively affect outcome measures such as click-through rates or engagement (Baltas 2003; Robinson et al. 2007). As such, we included several variables to control for different engagement by specific post characteristics. We included number of text characters (TEXT) to account for the length of the written message in posts. We included the number of exclamation marks (EXCL) to account for the message's emphasis in posts. We included the number of question marks (QUE) to account for the degree of SMI's call to action to their audience. We included the number of emojis (EMOJI) to account for the number of emotional expressions in posts. We included the number of mentions (MENTION) to account for the amount of referencing in posts. Finally, we included the number of hashtags (HASHTAG) to control for the number of search keywords used by SMIs in posts.

Third, users may find posts more attractive depending on type of content. It might be that users interact more with posts that contain dynamic content, such as videos, compared to static content, such as images, because they trigger more emotional reactions. Indeed, research has found that vivid content characteristics receive more likes and comments (De Vries, Gensler, and Leeflang 2012). Thus, we included

content type as another control variable in our model (CONTENT) to control for the type of content. CONTENT equals 1 if a post had dynamic content and 0 if it had static content. Table 3 shows the correlations of all variables collected for empirical analyses.

Estimation Strategy

The variance of our main dependent variable, ENGAGEMENT, exceeded its mean ($M = 1,192.41$, $VAR = 9,542,609$), indicating overdispersion in our data. We further conducted an overdispersion test (Cameron and Trivedi 1990) in a Poisson model and found strong evidence of overdispersion ($\alpha = 1,074.95$, $p < 0.001$). As a result, we estimated our main results with a negative binomial regression to profile overdispersion in count data (Greene 2007). Nonetheless, significance and direction of coefficients remained unchanged using a Poisson regression. The results of the Poisson regression appear in Table 4.

In our first step, we aimed to compare the engagement variables between sponsored posts and NSPs and estimated the results with the entire data set of 64,438 posts. In the second step, we aimed to compare different advertising appeals in sponsored posts (i.e., transformational versus informational). Therefore, we excluded NSPs from the sample. The effect of the type of advertising appeal in sponsored posts on engagement was estimated with a negative binomial regression with a subsample of 5,617 sponsored posts.

Moreover, our data may encounter heteroskedasticity. To test heteroscedasticity, we conducted a Breusch–Pagan test (Breusch and Pagan 1979) which showed that heteroskedasticity is present in the main sample of 64,438 posts ($BP = 4,767.4$, $DF = 13$, $p < 0.001$) and in the subsample of 5,617 sponsored posts ($BP = 255.52$, $DF = 13$, $p < 0.001$). Thus, we reported heteroskedasticity-consistent standard errors in all models. The negative binomial regression results

Table 4. Estimation results: Poisson regression.

Variables	Sponsored vs. Nonsponsored Posts			Informational vs. Transformational Posts		
	Likes	Comments	Engagement	Likes	Comments	Engagement
Dependent variables						
SPONSOR	0.102*** (0.018)	0.521*** (0.017)	0.112*** (0.017)			
FOL*SPONSOR	-0.051*** (0.011)	-0.187*** (0.013)	-0.053*** (0.011)			
INFO				0.066* (0.029)	0.024 (0.027)	0.064* (0.029)
FOL*INFO				-0.012 (0.021)	0.020 (0.018)	-0.011 (0.021)
Control variables						
FOL	0.966*** (0.010)	0.756*** (0.007)	0.962*** (0.010)	0.935*** (0.024)	0.500*** (0.016)	0.925*** (0.024)
FOL ²	-0.068*** (0.002)	-0.071*** (0.001)	-0.067*** (0.002)	-0.074*** (0.003)	-0.058*** (0.004)	-0.073*** (0.003)
TEXT	-0.032*** (0.008)	0.107*** (0.005)	-0.029*** (0.008)	-0.139*** (0.028)	-0.052*** (0.016)	-0.137*** (0.027)
EMOJI	-0.004 (0.007)	0.013* (0.006)	-0.004 (0.007)	0.030+ (0.017)	0.022* (0.010)	0.030+ (0.017)
MENTION	-0.055*** (0.009)	-0.106*** (0.007)	-0.056*** (0.008)	0.049*** (0.015)	-0.045*** (0.014)	0.047** (0.015)
HASHTAG	-0.050*** (0.007)	-0.131*** (0.006)	-0.051*** (0.007)	-0.064*** (0.015)	-0.119*** (0.014)	-0.065*** (0.015)
EXCL	-0.014+ (0.008)	-0.036*** (0.005)	-0.015+ (0.008)	0.013 (0.017)	0.017 (0.013)	0.013 (0.016)
QUE	-0.010+ (0.006)	0.061*** (0.005)	-0.009 (0.006)	0.019 (0.015)	0.056*** (0.011)	0.020 (0.015)
CONTENT	-0.085*** (0.024)	-0.145*** (0.019)	-0.085*** (0.024)	-0.250+ (0.144)	-0.604*** (0.105)	-0.255+ (0.142)
WORKHOUR	-0.001 (0.015)	-0.042*** (0.010)	-0.001 (0.015)	-0.012 (0.035)	0.032 (0.025)	-0.011 (0.034)
WEEKEND	-0.022 (0.015)	-0.039*** (0.011)	-0.022 (0.015)	-0.021 (0.045)	-0.075* (0.030)	-0.022 (0.044)
CONSTANT	6.691*** (0.009)	2.816*** (0.007)	6.711*** (0.008)	6.998*** (0.023)	3.488*** (0.021)	7.028*** (0.022)
Observations	64,438	64,438	64,438	5,617	5,617	5,617
AIC	51,518,475	1,491,897	51,837,171	4,637,446	170,622	4,658,114

Note. FOL was reported in 10,000 followers. We reported standardized coefficients and robust standard errors.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

appear in Table 5. We estimated the model using (1) number of likes, (2) number of comments, and (3) total number of likes and comments (i.e., engagement) as dependent variables.

Results

Descriptive Results

The study was conducted with 64,438 Instagram posts. The data contained sponsored posts and NSPs from 2015 to 2019. Of the posts, 99% were uploaded in 2018 and 2019; 43% during working hours; 24% over weekends; 72% contained emojis in their captions; 69% contained at least one hashtag; and 99% contained additional text characters such as question marks, exclamation marks, or regular words in their captions. The average number of total characters in a post was 254. A total of 8.37% of posts contained dynamic content.

The number of likes per post ranged between 0 and 111,386. The number of comments per post ranged between 0 and 150. The mean number of likes was 1,173.58 ($SD = 3,075.42$, $Mdn = 408$), and the mean number of comments was 18.82 ($SD = 26.88$, $Mdn = 7$). The total engagement per post ranged between 1 and 111,435. The mean number of engagements was 1,192.41 ($SD = 3,089.11$, $Mdn = 421$). The average number of followers was 27,807.95 ($SD = 56,154.07$, $Mdn = 10,305$).

The data contained 8.71% sponsored posts; of these, 39% had informational appeals. The mean number of engagements was 1,559.21 ($SD = 3,035.97$, $Mdn = 747$) for sponsored posts compared to 1,157.38 ($SD = 3,091.88$, $Mdn = 401$) for NSPs. The mean number of engagements was 1,481.38 ($SD = 2,750.60$, $Mdn = 675.5$) for sponsored posts with informational appeal compared to 1,609.18 ($SD = 3,205.19$, $Mdn = 800$) for those with transformational appeal.

Table 5. Estimation results: Negative binomial regression.

Variables	Sponsored vs. Nonsponsored Posts			Informational vs. Transformational Posts		
	Likes	Comments	Engagement	Likes	Comments	Engagement
Dependent variables						
SPONSOR	0.156*** (0.014)	0.556*** (0.017)	0.167*** (0.013)			
FOL*SPONSOR	-0.327*** (0.010)	-0.316*** (0.010)	-0.329*** (0.010)			
INFO				0.061* (0.027)	0.028 (0.029)	0.061* (0.026)
FOL*INFO				0.195*** (0.024)	0.053* (0.022)	0.187*** (0.024)
Control variables						
FOL	1.115*** (0.004)	0.657*** (0.004)	1.106*** (0.004)	0.849*** (0.013)	0.382*** (0.012)	0.838*** (0.013)
TEXT	0.010+ (0.006)	0.241*** (0.006)	0.014* (0.006)	-0.152*** (0.022)	-0.049** (0.016)	-0.149*** (0.021)
EMOJI	0.060*** (0.006)	0.076*** (0.005)	0.060*** (0.006)	0.044*** (0.012)	0.028* (0.011)	0.044*** (0.012)
MENTION	-0.010** (0.003)	-0.088*** (0.006)	-0.012*** (0.003)	-0.009 (0.012)	-0.070*** (0.014)	-0.011 (0.012)
HASHTAG	-0.092*** (0.005)	-0.197*** (0.006)	-0.094*** (0.005)	-0.023 (0.016)	-0.129*** (0.016)	-0.026+ (0.015)
EXCL	-0.054*** (0.005)	-0.056*** (0.006)	-0.053*** (0.005)	-0.008 (0.014)	0.033* (0.014)	-0.007 (0.014)
QUE	0.007+ (0.004)	0.117*** (0.005)	0.010* (0.004)	0.016 (0.012)	0.070*** (0.012)	0.018 (0.012)
CONTENT	-0.143*** (0.018)	-0.146*** (0.020)	-0.144*** (0.018)	-0.545*** (0.099)	-0.659*** (0.107)	-0.546*** (0.098)
WORKHOUR	-0.015+ (0.009)	-0.042*** (0.011)	-0.016+ (0.009)	0.014 (0.025)	0.081** (0.027)	0.016 (0.024)
WEEKEND	-0.025* (0.010)	-0.038** (0.012)	-0.025* (0.010)	-0.013 (0.029)	-0.094** (0.033)	-0.016 (0.0279)
CONSTANT	6.563*** (0.007)	2.722*** (0.008)	6.585*** (0.007)	6.894*** (0.019)	3.413*** (0.022)	6.926*** (0.019)
Observations	64,438	64,438	64,438	5,617	5,617	5,617
AIC	972,338	480,737	975,269	88,597	49,847	88,927

Note. FOL was reported in 10,000 followers. We reported standardized coefficients and robust standard errors.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Estimation Results

First, our results corroborated hypothesis 1—that sponsored posts receive more engagement (i.e., likes and comments) than NSPs. The coefficient of the variable SPONSOR was highly significant ($\beta = 0.167$, $p < 0.001$). Sponsored posts received more likes ($\beta = 0.156$, $p < 0.001$) and more comments ($\beta = 0.556$, $p < 0.001$) than nonsponsored ones. This indicated that users were more likely to like and comment on sponsored posts than NSPs; hypothesis 1 was supported. The fact that a post was sponsored positively influenced social media engagement.

Regarding public exposure, our results provided support for hypothesis 2—that having more followers negatively affected engagement with sponsored posts. The interaction coefficient (FOL*SPONSOR) was negative and highly significant ($\beta = -0.329$, $p < 0.001$). The results also showed that a higher number of followers negatively affected the number of likes ($\beta = -0.327$, $p < 0.001$) and comments ($\beta = -0.316$, $p < 0.001$). These results confirmed that social media

engagement with sponsored posts decreases with a higher number of followers.

Sponsored posts with informational appeals (INFO) positively affected engagement ($\beta = 0.061$) with a significance level of 5%. The link between informational appeals and number of likes was positive and significant ($\beta = 0.061$, $p < 0.05$), and the link between informational appeal and number of comments was positive and insignificant ($\beta = 0.028$, $p > 0.1$). There was no evidence that users significantly engaged more with sponsored posts with informational appeals; hypothesis 3 was not supported.

The results also showed that a higher number of followers positively influenced engagement with sponsored posts with informational appeals ($\beta = 0.187$, $p < 0.001$). The interaction coefficient (FOL*INFO) was positive and highly significant for number of likes ($\beta = 0.195$, $p < 0.001$) and positive for number of comments, with a significance level of 5% ($\beta = 0.053$). These findings supported hypothesis 4—that a higher number of followers positively affects social media engagement for sponsored posts with informational appeals.

Table 6. Estimation results: Linear regression.

Variables	Sponsored vs. Nonsponsored Posts			Informational vs. Transformational Posts		
	Likes	Comments	Engagement	Likes	Comments	Engagement
Dependent variables						
SPONSOR	2.540*** (0.302)	1.007*** (0.052)	2.749*** (0.303)			
FOL*SPONSOR	-2.158*** (0.147)	-0.145*** (0.025)	-2.192*** (0.148)			
INFO				-0.846 (0.650)	-0.079 (0.123)	-0.792 (0.652)
FOL*INFO				0.646* (0.318)	0.036 (0.060)	0.630* (0.319)
Control variables						
FOL	13.939*** (0.124)	2.219*** (0.021)	14.081*** (0.124)	11.581*** (0.502)	2.393*** (0.095)	11.736*** (0.503)
FOL ²	0.767*** (0.023)	-0.164*** (0.004)	0.748*** (0.023)	0.768*** (0.088)	-0.227*** (0.017)	0.741*** (0.088)
TEXT	-0.003*** (0.0003)	0.001*** (0.00005)	-0.003*** (0.0003)	-0.007*** (0.001)	-0.0002 (0.0001)	-0.007*** (0.001)
EMOJI	-0.008 (0.018)	0.015*** (0.003)	-0.005 (0.018)	0.263*** (0.060)	0.046*** (0.011)	0.265*** (0.060)
MENTION	0.116*** (0.026)	-0.058*** (0.004)	0.105*** (0.026)	0.171 (0.125)	-0.081*** (0.024)	0.155 (0.125)
HASHTAG	-0.046*** (0.006)	-0.029*** (0.001)	-0.050*** (0.006)	0.009 (0.022)	-0.028*** (0.004)	0.003 (0.022)
EXCL	-0.533*** (0.045)	-0.071*** (0.008)	-0.538*** (0.046)	-0.223 (0.145)	0.005 (0.028)	-0.211 (0.146)
QUE	-0.678*** (0.086)	0.288*** (0.015)	-0.622*** (0.087)	-0.016 (0.268)	0.197*** (0.051)	0.018 (0.268)
CONTENT	-2.223*** (0.175)	-0.272*** (0.030)	-2.260*** (0.176)	-6.031*** (1.061)	-1.380*** (0.201)	-6.180*** (1.064)
WORKHOUR	-0.503*** (0.097)	-0.136*** (0.017)	-0.519*** (0.097)	-0.617+ (0.358)	0.023 (0.068)	-0.610+ (0.359)
WEEKEND	-0.380*** (0.111)	-0.097*** (0.019)	-0.393*** (0.112)	-0.500 (0.420)	-0.253** (0.080)	-0.539 (0.421)
CONSTANT	7.961*** (0.134)	0.963*** (0.023)	8.103*** (0.134)	10.761*** (0.653)	2.117*** (0.124)	11.129*** (0.655)
Observations	64,438	64,438	64,438	5,617	5,617	5,617
R ²	0.700	0.391	0.700	0.665	0.293	0.664
Adjusted R ²	0.700	0.391	0.700	0.664	0.292	0.663

Note. FOL was reported in 10,000 followers. To reduce the skewness of the number of followers, we used a square root transformation. We reported standardized coefficients and robust standard errors.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Finally, we considered several control variables. We included several post characteristics in the models to control for any impacts of post characteristics (TEXT, EMOJI, MENTION, HASHTAG, EXCL, QUE). We also controlled for factors relating to a post's content (CONTENT) and time effects (WORKHOUR, WEEKEND). Because all control variables were mainly used to capture and control for a set of post-specific characteristics that may influence our dependent variables, no further interpretation was needed.

Robustness Checks

We conducted several robustness checks. In all robustness checks, we included the quadratic term of the number of followers on the Instagram profile to which the post was uploaded (FOL²) as an additional control variable. Adding the quadratic term allowed us to profile the decreasing engagement rate with an increasing number of followers.

First, a common alternative regression model is to transform the dependent variable and estimate a linear least-squares model (De Vries, Gensler, and Leeflang 2012; Sabate et al. 2014). We used a square root transformation to reduce the dependent variable's skewness from 12.05 to 3.06. We then ran a linear regression model (for the results, see Table 6). The results remained the same in direction and significance, except for the coefficient of INFO, which was negative and insignificant. Another alternative model was the zero-inflated regression model. All posts had at least one like or comment, which led us to exclude this regression estimation.

Second, we used a different subsample of the data set to compute the estimation results. We used a subsample that excluded extremely high and low engagement posts as an alternative data set. Posts may go viral (i.e., get proportionally more engagement) or may be unpopular (i.e., get proportionally less engagement) for various reasons; these extreme engagement

Table 7. Estimation results: Negative binomial regression with subsample.

Variables	Sponsored vs. Nonsponsored Posts			Informational vs. Transformational Posts		
	Likes	Comments	Engagement	Likes	Comments	Engagement
Dependent variables						
SPONSOR	0.044*** (0.011)	0.487*** (0.022)	0.060*** (0.010)			
FOL*SPONSOR	-0.069*** (0.012)	-0.061** (0.022)	-0.070*** (0.012)			
INFO				-0.024 (0.019)	0.077* (0.037)	-0.019 (0.019)
FOL*INFO				0.056*** (0.017)	-0.041 (0.036)	0.052** (0.016)
Control variables						
FOL	0.509*** (0.005)	0.377*** (0.009)	0.505*** (0.005)	0.505*** (0.015)	0.310*** (0.028)	0.495*** (0.015)
FOL ²	-0.043*** (0.002)	-0.035*** (0.002)	-0.043*** (0.001)	-0.076*** (0.005)	-0.040*** (0.007)	-0.074*** (0.005)
TEXT	-0.039*** (0.004)	0.229*** (0.007)	-0.033*** (0.004)	-0.081*** (0.013)	0.030 (0.022)	-0.075*** (0.013)
EMOJI	0.006+ (0.003)	0.046*** (0.006)	0.007* (0.003)	0.028** (0.009)	0.018 (0.015)	0.028** (0.009)
MENTION	0.012*** (0.003)	-0.094*** (0.007)	0.010*** (0.003)	-0.025* (0.010)	-0.108*** (0.020)	-0.027** (0.010)
HASHTAG	-0.003 (0.003)	-0.174*** (0.007)	-0.008* (0.003)	0.035** (0.011)	-0.181*** (0.023)	0.025* (0.011)
EXCL	-0.021*** (0.003)	-0.038*** (0.007)	-0.021*** (0.003)	-0.014 (0.011)	0.019 (0.018)	-0.014 (0.011)
QUE	-0.004 (0.003)	0.112*** (0.005)	-0.0004 (0.003)	0.016+ (0.009)	0.042* (0.017)	0.017+ (0.009)
CONTENT	-0.098*** (0.012)	-0.137*** (0.023)	-0.100*** (0.012)	-0.273*** (0.075)	-0.387** (0.136)	-0.277*** (0.073)
WORKHOUR	-0.011* (0.006)	-0.081*** (0.013)	-0.013* (0.006)	-0.011 (0.019)	0.067+ (0.037)	-0.009 (0.019)
WEEKEND	-0.012+ (0.007)	-0.029+ (0.015)	-0.013* (0.006)	-0.006 (0.022)	-0.093* (0.042)	-0.009 (0.021)
CONSTANT	6.232*** (0.004)	2.635*** (0.009)	6.260*** (0.004)	6.431*** (0.015)	3.222*** (0.029)	6.471*** (0.014)
Observations	38,674	38,674	38,674	3,211	3,211	3,211
AIC	529,709	280,527	531,406	44,865	27,135	45,042

Note. FOL was reported in 10,000 followers. We reported standardized coefficients and robust standard errors.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

posts may bias the estimation results. To exclude this possibility, we ran a negative binomial regression on a subsample, excluding the posts in the upper and lower 20% engagement quantiles. Again, the estimation results supported our hypotheses. Table 7 shows the regression results.

Third, we used an alternative dependent variable to estimate our results. A comment has a higher value than a like. Yoon et al. (2018) argued that writing a comment requires more effort than clicking the Like button. Comments allow companies to evaluate whether and how users perceive the advertised brands and products in a post, while likes do not (Lammenet 2019). Moreover, Kim and Yang (2017) stated that a comment has the same value as seven likes. Indeed, SMIs are paid up to five times as much for a comment than for a like. Based on this, we transformed the number of comments into likes by multiplying the number of comments by five. The alternative dependent variable, ENGAGEMENT, is the sum of number of likes and the number of comments

expressed in likes. The estimation results of a negative binomial regression appear in Table 8. The direction and significance of the variables of interest remained the same as in the main model. In sum, the estimates from a set of robustness checks offered further empirical evidence on the proposed econometric model.

Discussion

The influencer marketing industry on social media is growing. This study investigates social media engagement with sponsored posts on Instagram. By examining influencer marketing from an advertising perspective, this study has extended the knowledge of influencer marketing as an advertising strategy on Instagram. Furthermore, we investigated the role of SMIs' follower counts and advertising appeals on engagement with sponsored posts. The findings of this study suggest that SMIs' follower counts and the type of advertising appeal in sponsored posts affect social media engagement. Our findings respond to several

TABLE 8. Estimation results: Negative binomial regression with alternative dependent variables.

Variables Dependent variables	Sponsored vs. Nonsponsored Posts Engagement	Informational vs. Transformational Posts Engagement
SPONSOR	0.113*** (0.013)	
FOL*SPONSOR	-0.580*** (0.010)	
INFO		0.047+ (0.024)
FOL*INFO		0.083*** (0.022)
Control variables		
FOL	1.682*** (0.006)	1.184*** (0.016)
FOL ²	-0.097*** (0.001)	-0.127*** (0.005)
TEXT	0.013* (0.005)	-0.144*** (0.019)
EMOJI	0.065*** (0.006)	0.047*** (0.011)
MENTION	0.003 (0.003)	-0.011 (0.011)
HASHTAG	-0.083*** (0.005)	-0.025+ (0.014)
EXCL	-0.062*** (0.005)	-0.010 (0.013)
QUE	0.009* (0.004)	0.014 (0.011)
CONTENT	-0.205*** (0.016)	-0.546*** (0.084)
WORKHOUR	-0.038*** (0.008)	-0.015 (0.023)
WEEKEND	-0.030** (0.009)	-0.024 (0.027)
CONSTANT	6.773*** (0.006)	7.151*** (0.018)
Observations	64,438	5,617
AIC	981,953	89,553

Note. FOL was reported in 10,000 followers. We reported standardized coefficients and robust standard errors.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

calls for more research into influencer marketing (Carrillat and Ilicic 2019; Kees and Andrews 2019; Pentina, Guilloux, and Micu 2018; Voorveld 2019) in general, particularly on micro- and macroinfluencers (Voorveld 2019). We provide theoretical implications for researchers who aim to understand influencer marketing on social media. Moreover, we provide useful recommendations to advertisers and SMIs who wish to improve influencer marketing.

The first major finding concerns the relationship between the type of post (sponsored or nonsponsored) and social media engagement. This finding adds to the literature on influencer marketing as an advertising strategy on social media (Lou and Yuan 2019). Our results show that users engage more with sponsored posts than with NSPs. Our results suggest that consumers receive higher hedonic value from consuming sponsored posts, which positively affects social media engagement. In contrast to NSPs, SMIs are challenged to creatively embed brands and products in a compelling manner in sponsored posts. They spend more time, effort, and thought in sponsored

posts, resulting in richer content with higher hedonic value for social media users. This may imply that users receive higher hedonic value when consuming sponsored posts. This finding is in line with previous research that showed hedonic value can have an impact on post interaction and popularity (De Vries, Gensler, and Leeflang 2012; Hughes, Swaminathan, and Brooks 2019; Sabate et al. 2014).

Another important finding of the study relates to the effect of SMIs' follower counts on social media engagement with sponsored posts, which is in line with the findings of previous work (Voorveld 2019). We showed that the number of followers negatively affects social media engagement with sponsored posts. Despite macroinfluencers' sponsored posts being exposed to a larger number of users (De Veirman, Cauberghe, and Hudders 2017), people in larger groups feel less responsible and addressed and then stand by rather than act (Darley and Latané 1968). It is not surprising to observe that engagement with sponsored posts decreases with a higher number of followers; this may be because users passively engage

(i.e., looking effect) rather than actively engage (i.e., liking and commenting) with macroinfluencers' sponsored posts.

We do not find any empirical evidence that the type of advertising appeal in sponsored posts affects social media engagement. We find no support that users engage significantly more with sponsored posts with informational appeals compared to transformational appeals. One potential explanation for this finding might be that users give first or even more attention to the SMIs' follower counts rather than the type of advertising appeal when evaluating sponsored posts. When being exposed to sponsored posts, users might first evaluate the source (i.e., macroinfluencers versus microinfluencers), which then influences how they evaluate advertising messages in sponsored posts. This is in line with our findings that show users engage more with sponsored posts with informational appeals by macroinfluencers compared to microinfluencers.

Moreover, our study offers a third major finding on the relationship of the type of advertising appeal in sponsored posts and SMIs' follower counts. Our results showed that users engage more with sponsored posts with informational appeals by macroinfluencers than by microinfluencers. We argue that users evaluate advertising messages in sponsored posts followed by more people as more interesting and relevant, and thus pay more attention to it compared to microinfluencers. As such, users evaluate advertising messages by macroinfluencers more carefully. In line with the predictions of the ELM, informational appeals in sponsored posts by macroinfluencers exert a greater impact on persuasion.

Finally, this study investigated factors that determine the effectiveness of influencer marketing on Instagram. Previous work on influencer marketing has focused on understanding SMIs as a new endorser type (Hearn and Schoenhoff 2016; Jin, Muqaddam, and Ryu 2019; Schouten, Janssen, and Verspaget 2020), SMI-brand fit (Breves et al. 2019; e.g., Kamins 1990), or the effects of SMIs' advertising on consumers' attitudes and intentions toward brands and products (e.g., De Jans, Cauberghe, and Hudders 2018; Evans, Hoy, and Childers, Lemon, and Hoy 2018). In contrast, we have increased the knowledge of whether and how SMIs' follower counts and the type of advertising appeal affect social media engagement with sponsored posts. With this variety, we have extended and refined previous findings on influencer marketing's effectiveness (Hughes, Swaminathan, and Brooks 2019).

Theoretical Implications

From a theoretical perspective, our study implies that both content and SMIs' characteristics explain how social media users engage with sponsored posts. For sponsored posts to be effective in engaging social media users, they have to contain different advertising appeals depending on the public exposure of SMIs. This study tested not only if influencer marketing is a promising advertising strategy for both advertisers and SMIs but also if and how content characteristics (i.e., advertising appeals) and SMI characteristics (i.e., number of followers) affect the effectiveness of influencer marketing (i.e., social media engagement).

In a broader sense, we extend the literature on information processing theory (Cacioppo and Petty 1979; Petty, Cacioppo, and Heesacker 1981; Petty, Cacioppo, and Schumann 1983). We find that a higher number of followers positively affects social media engagement with informational appeals compared to transformational appeals in sponsored posts. This result suggests that users engage more with sponsored posts with informational appeals by macroinfluencers compared to microinfluencers. The reason for this finding can be found in information processing theory. In line with the predictions of the ELM, we argue that users evaluate sponsored posts followed by more people as more interesting and relevant and therefore pay more attention to the message. Hence, users evaluate macroinfluencers' sponsored posts with informational appeals more carefully compared to those by microinfluencers, which results in higher engagement. This study is a starting point for future research investigating how users evaluate advertising appeals in sponsored posts by micro- and macroinfluencers.

Although many studies in the literature of effectiveness of influencer marketing focus on consumer attitudes (Lou and Yuan, 2019; van Reijmersdal et al. 2020; Yuan and Lou 2020), it is important to take into account behavioral and financial aspects as well. Our study extends prior research on the effectiveness of influencer marketing by investigating social media engagement, which presents an important KPI for influencer marketing that accounts for both the behavioral and the financial outcomes. Our findings show that users engage more with sponsored posts than NSPs, particularly for those posts by microinfluencers. Our research thus contributes to prior research by highlighting the importance of sponsored posts for both advertisers and SMIs alike, and in showing that the effectiveness derived from sponsored

posts appears greater for microinfluencers than macroinfluencers.

While previous research focuses on SMI characteristics influencing effectiveness (Erdogan 1999; Ohanian 1990), the current study suggests that both content and SMI characteristics are indispensable for effective influencer marketing. Our research shows that users engage more with sponsored posts with informational appeals, particularly by macroinfluencers. Specifically, our findings build on prior research by examining both content and source characteristics in the context of influencer marketing. Hence, this study builds a groundwork for future empirical research in understanding the interplay of content and source characteristics for successful influencer marketing.

Managerial Implications

One important question in the current debate of the effectiveness of influencer marketing is how to increase social media engagement with sponsored posts. Our study has some useful recommendations for advertisers interested in influencer marketing.

Because social media users engage differently with sponsored posts by micro- compared to macroinfluencers, advertisers might place more importance on selecting SMIs depending on their advertising strategy. Specifically, advertisers are recommended to cooperate with microinfluencers in sponsored posts when aiming for long-term consumer actions. Moreover, advertisers should consider macroinfluencers when aiming for short-term consumer actions. In such cases, informational appeals in sponsored posts might be preferred to transformational appeals. Because users engage more with sponsored posts with informational appeals by macroinfluencers, microinfluencers are a better fit when considering transformational appeals in sponsored posts. Finally, advertisers should ensure that SMIs produce and upload sponsored posts of high hedonic value independent of their advertising appeal.

This study conveys some meaningful recommendations to SMIs. Users engage more with sponsored posts than with NSPs. SMIs spend more time, effort, and thought to produce sponsored posts because they receive something in return from the company. Yet SMIs are recommended to signal trust and expertise to their followers independent of the collaboration. They can opt for producing NSPs of equal hedonic value, which could positively shape their relationships with their followers.

Limitations and Avenues for Future Research

This study has several limitations. First, while we believe that social media engagement is one of the most important KPIs in influencer marketing, we recognize other relevant KPIs that were not included in our research are worthwhile to investigate. For example, the sentiment of social media engagement, conversion, or click rates might further improve the understanding of social media engagement with sponsored posts. Future research could test the effect of sponsored posts and their advertising appeals on different KPIs to explore whether a particular advertising appeal would be effective in inducing an expected outcome.

Second, we focused on comparing social media engagement between sponsored posts and NSPs. We excluded unpaid sponsored posts from this study. Going a step further, future research is encouraged to extend our findings by building and training a classifier to detect paid versus unpaid sponsored as well as nonsponsored that allows a deeper understanding of how users engage with different type of posts on Instagram.

Third, our study operationalized a dichotomous classification of the advertising appeals in sponsored posts with either informational or transformational appeals. However, sponsored posts might contain both types of appeals (Puto and Wells 1984). In future research, scholars could build on our research by investigating whether and how social media engagement varies for sponsored posts with a more complex classification of advertising appeals. For example, future researchers could build and train a classifier for advertising appeals accounting for all combinations of information and transformational appeals in sponsored posts.

Fourth, one primary goal of the study was to explore how the size of the follower count of SMIs affect users' engagement with sponsored posts and their advertising appeals. We limited the investigation of the characteristics of the source on the number of followers (i.e., microinfluencers versus macroinfluencers). However, gender of the SMIs is another characteristic that might confound results. Female users might be more likely to build parasocial interaction with female SMIs, while male users might be more likely to build parasocial interaction with male SMIs (Hudders and Jans 2022). Future research can continue this line of inquiry by exploring how the SMI gender affects the efficiency of sponsored posts and their advertising appeals.

Next, another primary goal of the study was to compare social media engagement between sponsored posts and NSPs. Findings from our prestudy suggest that heightened hedonic value in sponsored posts might be a potential explanation for higher social media engagement compared to NSPs. However, these findings are based on a qualitative study, which does not allow us to make causal conclusions. We encourage future research to test hedonic value as a potential moderator/mediator in a controlled experimental setting.

Further, we suggest that a potential explanation for the positive effects of the number of followers on engagement with sponsored posts with informational appeals can be found in information processing theory. Investigating the underlying mechanism of how consumers elaborate the different type of advertising messages by micro- and macroinfluencers could further increase understanding of both theoretical and managerial perspectives.

Finally, we analyzed 64,438 Instagram posts. To achieve this breadth of Instagram data, we had to accept some limitations in our data and empirical setup. Some posts included incomplete information due to data extraction, data privacy, and security. Nonetheless, we carefully prepared, cleaned, and tested the data to provide a reliable data set of highest quality. We demonstrated the reliability of the measurements and robustness of the abovementioned effects. While we are confident that the results remain unaffected, a small risk of unobserved heterogeneity persists.

Conclusion

This study responded to the call for more research into influencer marketing (Voorveld 2019) by exploring social media engagement with sponsored posts. The unique contribution of this research is to address the research objectives with a data-driven approach. Rather than focusing on consumers' attitudes and intentions, we have taken an initial step in exploring users' actual social media behaviors with sponsored posts and NSPs on Instagram. This study is among the first in understanding the determinants affecting social media engagement with sponsored posts on Instagram. We trust that future research will investigate further determinants (e.g., product categories or SMI-brand fit) beyond SMI follower counts and advertising messages, allowing us to deepen the understanding of social media engagement with sponsored posts.

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Disclosure Statement

No potential conflicts of interest were reported by the authors.

ORCID

Jana Gross  <http://orcid.org/0000-0002-9355-8761>
 Florian von Wangenheim  <http://orcid.org/0000-0003-3964-2353>

Data Availability Statement

Due to the nature of this research, participants of this study did not agree for their data to be shared publicly, so supporting data are not available.

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